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Visual Analyst: Data Tools Capability Report

Executive Summary

This report demonstrates the Visual Analyst's data tools by querying 15 authoritative public APIs and rendering live data as charts. Every number is sourced in real-time with zero API keys required. Sources span economics (World Bank, Treasury, Census), policy (Federal Register, SEC), health (WHO), science (NCBI, EMBL-EBI, CERN), environment (USGS, NWS/NOAA, GBIF), and space (NASA SVS).

1. World Bank: Top Economies by GDP

Gross Domestic Product for major economies in 2023, measured in trillions of US dollars. Data sourced from the World Bank Open Data API (indicator NY.GDP.MKTP.CD).

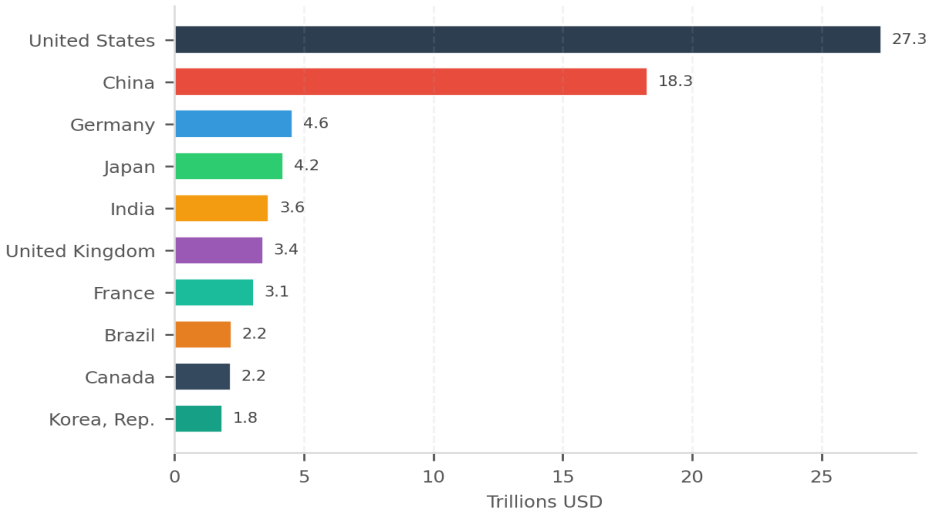

Figure 1. GDP by Country (2023) — Trillions USD
Source: World Bank Open Data API — Indicator NY.GDP.MKTP.CD

2. World Bank: Life Expectancy Trends (2015-2022)

Life expectancy at birth for 5 countries from 2015 to 2022. Notice the COVID-19 impact visible in 2020-2021 dips across most nations.

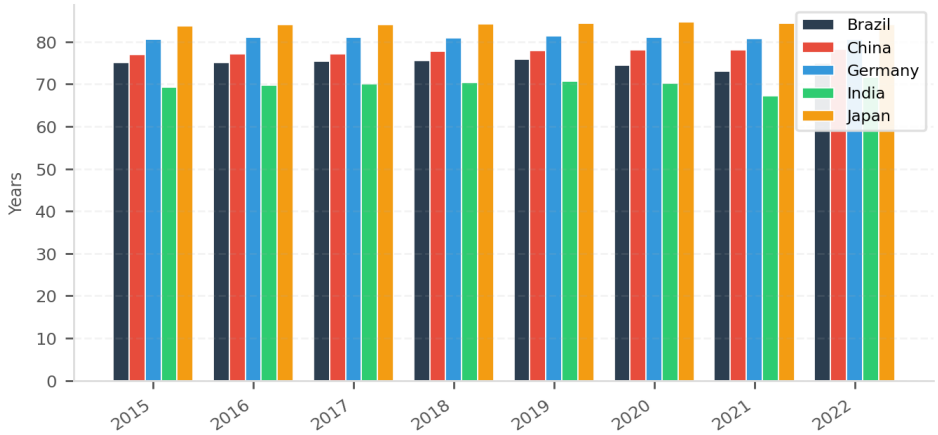

Figure 2. Life Expectancy at Birth (Years)

Source: World Bank Open Data API — Indicator SP.DYN.LE00.IN

3. WHO: Global Life Expectancy — Highest vs Lowest

The 5 countries with highest and 5 with lowest life expectancy (both sexes). The gap between the top and bottom is 0.0 years.

4. SEC EDGAR: Apple Inc. Filing Distribution

Distribution of Apple's most recent 20 SEC filings by form type. Most common: 4 (15 filings).

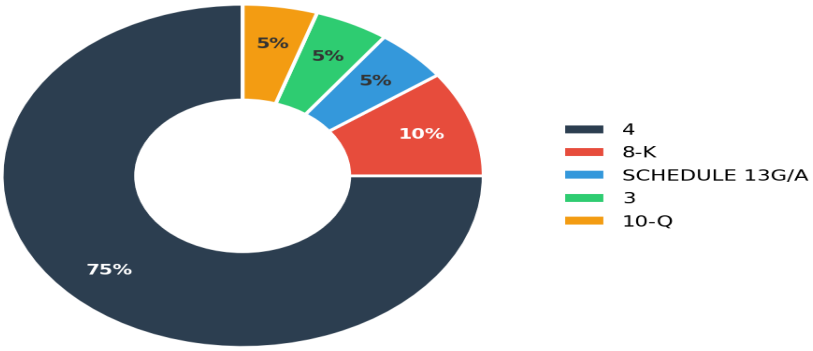

Figure 3. Apple SEC Filing Types (Recent 20)

Source: SEC EDGAR — CIK 0000320193 (Apple Inc.)

5. Federal Register: AI Policy Documents

Classification of 20 recent federal documents mentioning 'artificial intelligence' by document type.

Recent documents: - Ocean Research Advisory Panel (ORAP) (2026-03-30) - Risk-Informed, Technology-Inclusive Regulatory Framework for Advanced (2026-03-30) - Improving Wage Protections for the Temporary and Permanent Employment (2026-03-27) - Notice Pursuant to the National Cooperative Research and Production Ac (2026-03-26) - General and Plastic Surgery Devices; Reclassification of Optical Diagn (2026-03-25)

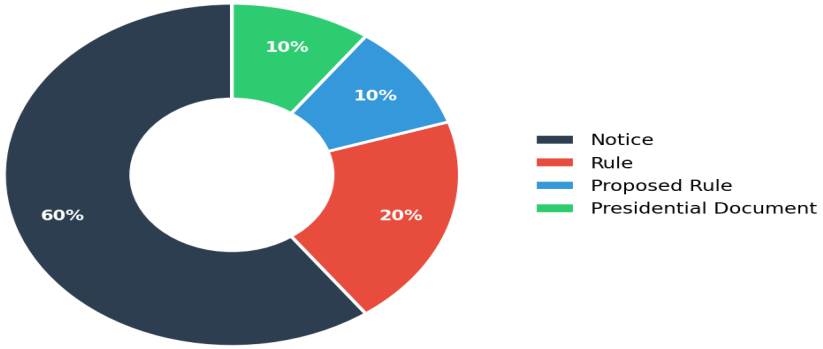

Figure 4. AI-Related Federal Documents by Type

Source: Federal Register API — federalregister.gov

6. U.S. Treasury: Exchange Rates (USD per Unit)

Current exchange rates from the U.S. Treasury for 12 major currencies. Values represent units of foreign currency per 1 USD.

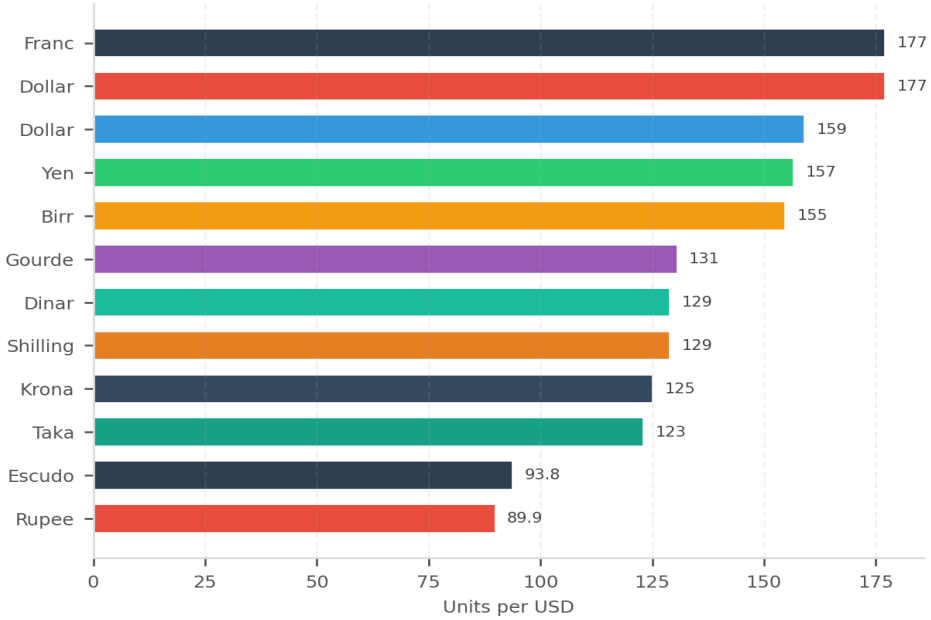

Figure 5. Exchange Rates (Foreign Currency per 1 USD)

Source: U.S. Treasury Fiscal Data — Rates of Exchange

7. U.S. Census: Median Household Income (Top 15 States by Pop.)

Median household income for the 15 most populous U.S. states (ACS 2023). Highest income: Massachusetts ($101,341).

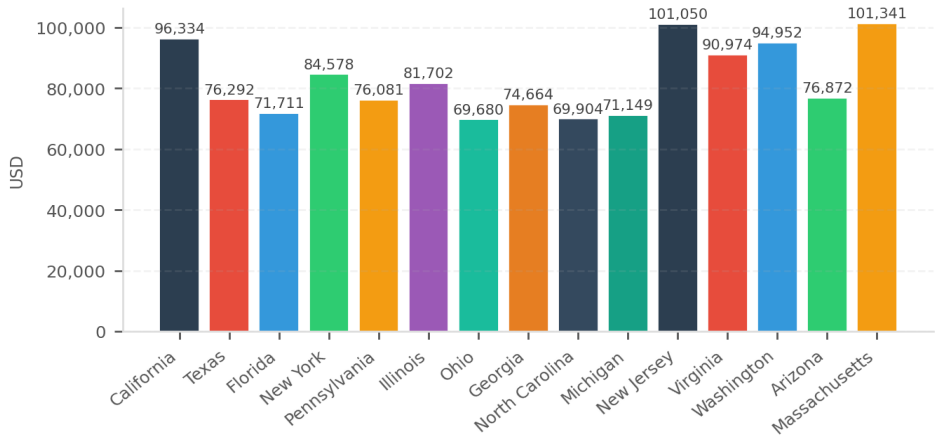

Figure 6. Median Household Income by State (USD)

Source: U.S. Census Bureau — American Community Survey 2023 (Table B19013)

8. NCBI: Cancer Gene Genomic Spans

Genomic span (kilobases) of 8 key cancer-associated genes. Larger spans often correlate with complex regulation.

Genes: EGFR (chr7, 192.6 kb), BRCA1 (chr17, 126.0 kb), PTEN (chr10, 108.3 kb), PIK3CA (chr3, 92.0 kb), KRAS (chr12, 45.7 kb), AKT1 (chr14, 26.4 kb), TP53 (chr17, 19.1 kb), MYC (chr8, 7.5 kb)

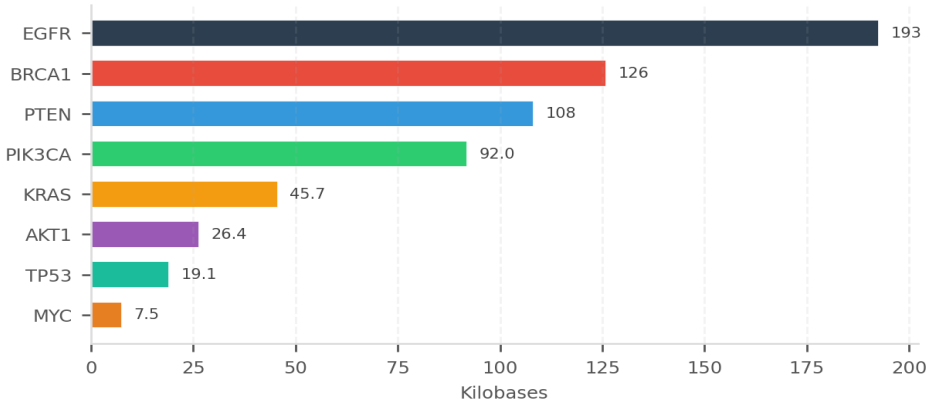

Figure 7. Gene Genomic Span (Kilobases)

Source: NCBI Datasets API — Gene records (Homo sapiens)

9. USGS: Potomac River Streamflow (7-Day Trend)

Instantaneous discharge at the Potomac River near Washington DC over 7 days. Peak: 10,800 cfs, Low: 8,250 cfs. Total readings: 1,955.

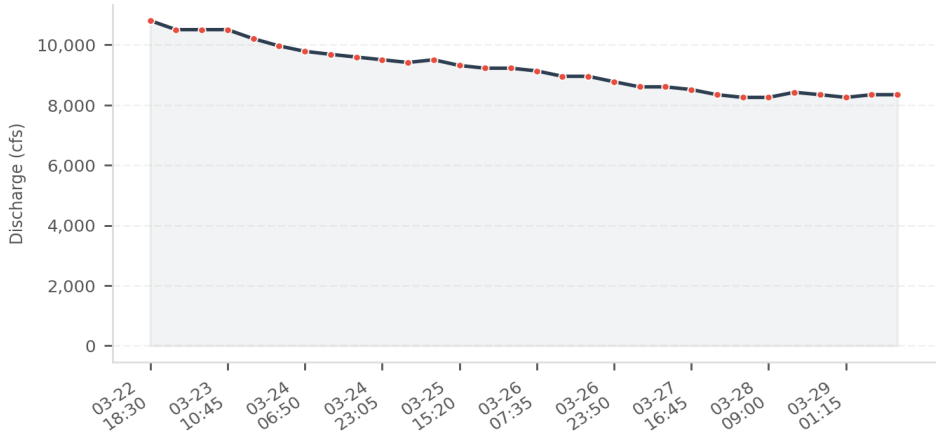

Figure 8. Potomac River Discharge (cubic ft/s)

Source: USGS Water Services — Site 01646500, Parameter 00060 (Discharge)

10. GBIF: Tiger (Panthera tigris) Occurrence by Country

Geographic distribution of 300 tiger occurrence records from the Global Biodiversity Information Facility. Total records globally: 7,762.

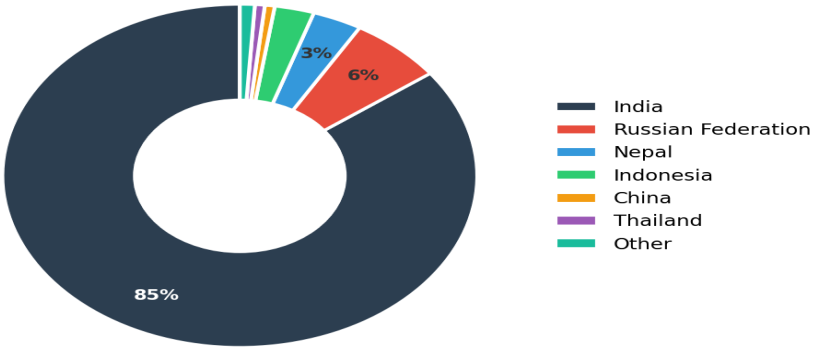

Figure 9. Tiger Occurrence Records by Country

Source: Global Biodiversity Information Facility (GBIF) — Panthera tigris

11. NWS/NOAA: JFK Airport — Temperature & Wind (24h)

Hourly observations at JFK International Airport. Temperature range: 7.0°C to 9.0°C. Max wind: 43 km/h.

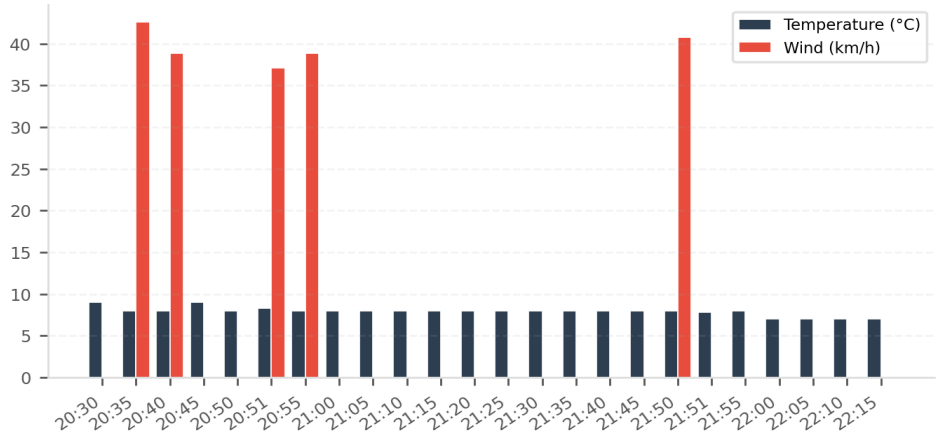

Figure 10. JFK Airport: Temperature (°C) & Wind Speed (km/h)

Source: National Weather Service API — Station KJFK

12. EMBL-EBI: SARS-CoV-2 PDB Structures by Taxonomy

Distribution of 6202 SARS-CoV-2 related protein structures in the Protein Data Bank, grouped by source organism taxonomy ID.

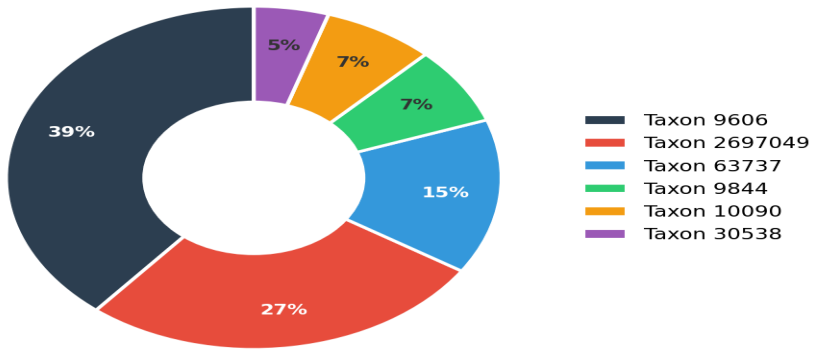

Figure 11. SARS-CoV-2 PDB Structures by Taxonomy (6202 total)

Source: EMBL-EBI Search — Protein Data Bank (PDB)

14. NASA SVS: Earth & Climate Visualizations Over Time

Scientific visualizations from NASA's SVS related to Earth and climate science, grouped by release year.

Recent visualizations: - Arctic Sea Ice Maximum 2026 - Sea Level Through a Porthole (2026) - GRACE FO Soil Moisture Within Continental United States: Monitori - The Earth System Science Spheres - Deserts of Africa and the Middle East

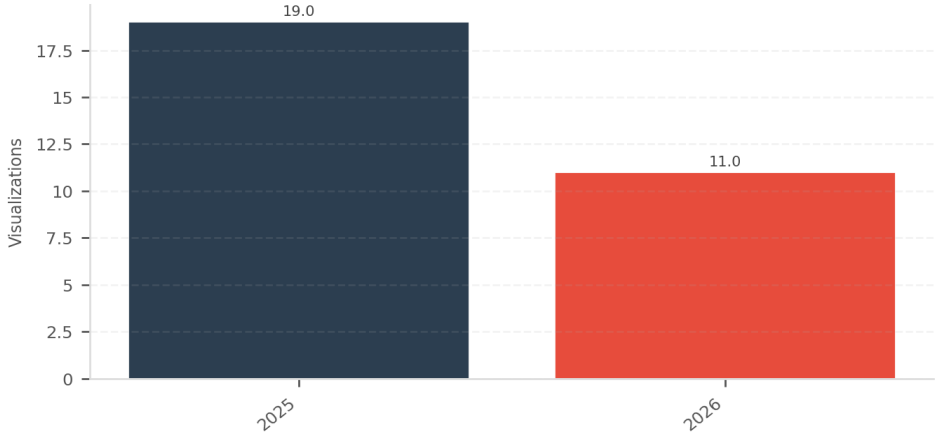

Figure 12. NASA Climate Visualizations by Year

Source: NASA Scientific Visualization Studio — svs.gsfc.nasa.gov

Sources and References

No specific sources were cited in the text.